

Advanced (Habanero)

Q1. Factor the expression: $x^2 + 5x + 4x + 20$

- (A) $x(x + 5) + 4(x + 5)$
- (B) $x(x + 4) + 5(x + 4)$
- (C) $(x + 4)(x + 5)$
- (D) $(x + 5)(x + 4)$
- (E) $x^2 + 9x + 20$

Q2. Factor the expression: $x^2 + 7x + 2x + 14$

- (A) $x(x + 7) + 2(x + 7)$
- (B) $x(x + 2) + 7(x + 2)$
- (C) $(x + 7)(x + 2)$
- (D) $(x + 2)(x + 7)$
- (E) $x^2 + 9x + 14$

Q3. Factor the expression: $y^2 + 4y + 3y + 12$

- (A) $y(y + 4) + 3(y + 4)$
- (B) $y(y + 3) + 4(y + 3)$
- (C) $(y + 4)(y + 3)$
- (D) $(y + 3)(y + 4)$
- (E) $y^2 + 7y + 12$

Q4. Factor the expression: $x^2 + 6x + 5x + 30$

- (A) $x(x + 6) + 5(x + 6)$
- (B) $x(x + 5) + 6(x + 5)$
- (C) $(x + 6)(x + 5)$
- (D) $(x + 5)(x + 6)$
- (E) $x^2 + 11x + 30$

Q5. Verify the factoring of the expression: $(x + 3)(x + 4) =$

- (A) $x^2 + 3x + 4x + 7$
- (B) $x^2 + 3x + 4x + 12$
- (C) $x^2 + 7x + 12$
- (D) $x^2 + 3x + 4x + 14$
- (E) $x^2 + 7x + 14$

Q6. Verify the factoring of the expression: $(y + 2)(y + 5) =$

- (A) $y^2 + 2y + 5y + 7$
- (B) $y^2 + 2y + 5y + 10$
- (C) $y^2 + 7y + 10$
- (D) $y^2 + 2y + 5y + 12$
- (E) $y^2 + 7y + 12$

Q7. Factor the expression: $z^2 + 8z + 6z + 48$

- (A) $z(z + 8) + 6(z + 8)$
- (B) $z(z + 6) + 8(z + 6)$
- (C) $(z + 8)(z + 6)$
- (D) $(z + 6)(z + 8)$
- (E) $z^2 + 14z + 48$

Q8. Factor the expression: $x^2 + 9x + 3x + 27$

- (A) $x(x + 9) + 3(x + 9)$
- (B) $x(x + 3) + 9(x + 3)$
- (C) $(x + 9)(x + 3)$
- (D) $(x + 3)(x + 9)$
- (E) $x^2 + 12x + 27$

Open Ended Questions (FRQ)

Q9. Factor the expression: $2x^2 + 5x + 2x + 10$

Q10. Verify the factoring of the expression: $(x + 2)(x + 6) =$ Using the factored form, find the product of $(x + 2)$ and $(x + 6)$ when $x = 3$

Chef's Tips

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- Tip 1: Use the distributive property to check the factoring of expressions.
- Tip 2: Pay attention to the signs and coefficients when factoring and verifying expressions.
- Tip 3: Make sure students provide the correct factored form and verify it by multiplying the factors.